

Haifa Capital (K) Series – Construction and Interpretation Report

This report explains how the **Haifa capital (K) series** used in the thesis are constructed from raw financial data, and how to interpret each component of the final **monthly cluster K panel** for Haifa. It is written so it can be dropped into a separate chat (or into the thesis text) to make the data work fully transparent.

The key objects are:

- Operator-level K for **Haifa Port Company (HPC)** – the legacy operator.
- Landlord-level K for **Israel Ports Company (IPC, Haifa cluster)** – the infrastructure owner.
- The **Haifa cluster K**, which is simply the sum of HPC and IPC capital.
- For each of these, **three depreciation scenarios** – `low`, `central`, and `high` – corresponding to different depreciation rates δ .

All series are expressed in **real thousands of NIS, rebased to 2019 prices**, and are available at **monthly frequency** from **December 2018 through December 2024**.

1. Overview: What the final Haifa K panel contains

The main output is a file (and corresponding DataFrame) called:

```
20_K_cluster_Haifa_HPC_IPC_monthly.tsv
```

This file is in **long panel format**. Each row is a particular **entity-month** observation, where:

- `port` = "Haifa" for all rows.
- `company` takes three values:
 - "Haifa Port Company (legacy)" – operator K at Haifa.
 - "Israel Ports Company (Haifa cluster)" – landlord K at Haifa.
 - "Haifa cluster (HPC + IPC)" – total cluster K.
- `operator_or_owner` describes the role of each company (legacy operator, landlord, or cluster aggregate).
- `month` is a timestamp (first day of the month).
- `year` is the calendar year extracted from `month`.

For each row, the file contains three **scenario-specific K series**:

- `K_low` – capital stock under a **low depreciation rate** δ (slow depreciation, higher K).
- `K_central` – capital stock under the **central / baseline** δ (your preferred benchmark).
- `K_high` – capital stock under a **high depreciation rate** δ (fast depreciation, lower K).

Additionally, the file keeps **decomposition columns** that allow you to see how the cluster total is built from operator and landlord pieces:

- `K_HPC_low`, `K_HPC_central`, `K_HPC_high` – HPC capital by scenario.
- `K_IPC_low`, `K_IPC_central`, `K_IPC_high` – IPC capital by scenario.
- `K_cluster_low`, `K_cluster_central`, `K_cluster_high` – the sum of HPC and IPC capital by scenario.

By construction, for each month and scenario:

$$K_{\text{cluster},s}(t) \equiv K_{\text{HPC},s}(t) + K_{\text{IPC},s}(t),$$

for each scenario $s \in \{\text{low, central, high}\}$.

The panel covers **73 months** – one row for each month from **2018-12-01 to 2024-12-01** – and for each month you have three company rows (HPC, IPC, cluster).

2. From raw accounts to real annual K for each company

The monthly Haifa K panel is built on top of a common **three-step architecture**, applied separately to **HPC** and **IPC**:

1. **Step 1 – Raw nominal accounts → real annual financials**
2. **Step 2 – Annual Perpetual Inventory Method (PIM)**
3. **Step 3 – Monthly K (Path A or Path B)**

2.1 Step 1 – Deflating raw PPE and investment flows

For each company, you start from hand-built annual tables extracted from their financial statements:

- **HPC** (Haifa Port Company, legacy operator):
 - You extract net property, plant, and equipment (PPE) from the balance sheet and PPE notes, along with additions, disposals, and accumulated depreciation.
 - You also extract depreciation & amortization expense and cash-flow capex lines from the cash-flow statement.
- **IPC (Haifa cluster):**
 - You extract a Haifa-specific PPE table (by class) and a cash-flow style investment table, which together give total Haifa port-cluster PPE, additions, and disposals on IPC's books.

These nominal quantities are then merged with an **annual capital deflator** based on the OECD GFCF price index for Israel, rebased so that **2019 = 1**. This yields, for each calendar year and each company:

- `ppe_net_nom` → `K_book_real` (real net PPE at year-end)

- `additions_nom` → `I_real` (real gross investment)
- `depr_nom` → `depr_real` (real depreciation expense)
- `disposals_nom` → `disposals_real` (real disposals / retirements)

Everything is expressed in **2019-NIS thousands**, so that stocks and flows are comparable across time and across companies.

At this stage you also define **depreciation scenarios** – usually 4% (low), 6% (central), and 8% (high) – in a small JSON file. These are not applied yet in Step 1, but they are used in the PIM recursion in Step 2.

2.2 Step 2 – Annual PIM for multiple depreciation scenarios

In Step 2, you run a standard **Perpetual Inventory Method (PIM)** to construct capital stocks under alternative depreciation assumptions.

For each company (HPC, IPC) and each scenario s with depreciation rate δ_s , you define:

- An initial stock anchor in the first year:

$$K_1^{\text{PIM},s} = K_{\text{book, real},1}$$
- A PIM recursion for subsequent years t :

$$K_t^{\text{PIM},s} = (1 - \delta_s)^{\text{gap}_t} K_{t-1}^{\text{PIM},s} + I_t - D_t,$$

where I_t is real gross investment, D_t is real disposals, and `gap_t` is the number of years between observations (usually 1).

This yields, for each company and year:

- `K_PIM_real_low` – annual real capital stock under low δ .
- `K_PIM_real_central` – stock under central/baseline δ .
- `K_PIM_real_high` – stock under high δ .

You also compute quality-control gaps vs the accounting net PPE (`K_book_real`) to check that the PIM series stays anchored in the actual accounts.

2.3 Step 3 for IPC – Smooth monthly PIM (Path A)

For **IPC**, you construct a **smooth monthly capital path (Path A)** directly from the annual PIM series:

- For each year, the December PIM stock is treated as an anchor.
- Between December anchors, you interpolate the log of capital over months and exponentiate to get a smooth monthly series.

For each scenario s , this yields monthly series:

- `K_PIM_lin_low` – monthly IPC K under low δ .
- `K_PIM_lin_central` – monthly IPC K under central δ .
- `K_PIM_lin_high` – monthly IPC K under high δ .

The Haifa IPC monthly file `12_K_B_monthly_Haifa_IPC_PIM_lin.tsv` thus contains, for each month, `K_PIM_lin_low`, `K_PIM_lin_central`, and `K_PIM_lin_high`, all in 2019-NIS thousands.

2.4 Step 3 for HPC – Lumpy Path B K plus scenarios

For **HPC**, you construct a richer **Path B** series that combines:

1. A **lumpy “big project” module** tracking major capital projects (e.g. cranes, marine works) with project-level commissioning dates and lifetimes.
2. A **smooth background capital** component calibrated so that, for each depreciation scenario, the sum of big projects and background equals the annual PIM stock.

Concretely:

1. You build a project table `Haifa_big_projects.csv` that lists each major HPC project, its commissioning date, an estimated asset life, and its real cost in thousands of NIS.
2. For each project p , you build a monthly capital path $K_{p,t}$ that is zero before commissioning, jumps at the commissioning month, and then decays geometrically at a project-specific monthly depreciation rate.
3. Summing across projects gives the **big-project capital**:

$$K_{\text{big},t} = \sum_p K_{p,t}.$$

1. For each year y and each PIM scenario s , you then define the **annual background capital** $K_{\text{bg},y}^s$ so that:

$$K_{\text{PIM},y}^s = K_{\text{big},y} + K_{\text{bg},y}^s,$$

where $K_{\text{PIM},y}^s$ is the annual PIM stock from Step 2 (scenario s) and $K_{\text{big},y}$ is the December-year value of the big-project capital.

1. You interpolate $K_{\text{bg},y}^s$ over months (again log-linearly) to build **monthly background capital paths** $K_{\text{bg},t}^s$.
2. Finally, you define the **Path B total capital** for each scenario as:

$$K_{\text{PathB},t}^s = K_{\text{big},t} + K_{\text{bg},t}^s.$$

The resulting monthly file `03_K_B_monthly_Haifa_PathB.tsv` contains, for HPC and each month:

- `K_PathB_low`, `K_PathB_central`, `K_PathB_high` – Path B capital under each δ .
- `K_bg_low`, `K_bg_central`, `K_bg_high` – background capital under each δ .
- `K_big_central` – the scenario-invariant big-project component.

These are again all in 2019-NIS thousands.

3. Merging HPC and IPC into a Haifa cluster K panel

The file `20_K_cluster_Haifa_HPC_IPC_monthly.tsv` is built by merging:

- **HPC Path B** monthly K:
 - `03_K_B_monthly_Haifa_PathB.tsv` (HPC)
- **IPC PIM-lin** monthly K:
 - `12_K_B_monthly_Haifa_IPC_PIM_lin.tsv` (IPC)

The merge is done on:

- `port` (Haifa),
- `month`,
- `year`.

3.1 Scenario-specific HPC and IPC K

For each month, the script renames the scenario-specific K columns:

- For HPC:
 - `K_PathB_low` → `K_HPC_low`
 - `K_PathB_central` → `K_HPC_central`
 - `K_PathB_high` → `K_HPC_high`
- For IPC:
 - `K_PIM_lin_low` → `K_IPC_low`
 - `K_PIM_lin_central` → `K_IPC_central`
 - `K_PIM_lin_high` → `K_IPC_high`

So, after the merge, each month has six key K columns representing HPC and IPC capital separately under all three scenarios.

3.2 Building cluster K for each scenario

For each month and each scenario s , the cluster capital is defined as the sum of HPC and IPC capital:

$$K_{\text{cluster},s}(t) = K_{\text{HPC},s}(t) + K_{\text{IPC},s}(t).$$

In the data, this is implemented as:

- `K_cluster_low` = `K_HPC_low` + `K_IPC_low`
- `K_cluster_central` = `K_HPC_central` + `K_IPC_central`
- `K_cluster_high` = `K_HPC_high` + `K_IPC_high`

Because all components are in the same units (real thousands of 2019-NIS), the sums are directly meaningful.

3.3 Constructing the long panel with three company values

To make the panel convenient for econometrics, the script creates **three sets of rows** for each month:

1. **HPC rows** – operator K
2. `company = "Haifa Port Company (legacy)"`
3. `operator_or_owner = "Haifa Port Company (legacy operator)"`
4. For each scenario s :
 - `K_s = K_HPC_s`
5. **IPC rows** – landlord K
6. `company = "Israel Ports Company (Haifa cluster)"`
7. `operator_or_owner = "Israel Ports Company (landlord / infrastructure owner)"`
8. For each scenario s :
 - `K_s = K_IPC_s`
9. **Cluster rows** – total cluster K
10. `company = "Haifa cluster (HPC + IPC)"`
11. `operator_or_owner = "Haifa port cluster (HPC + IPC)"`
12. For each scenario s :
 - `K_s = K_cluster_s`

All three blocks also retain the decomposition columns `K_HPC_*`, `K_IPC_*`, and `K_cluster_*`, so that even when you subset the panel to `company == "Haifa cluster (HPC + IPC)"`, you can still see how the cluster capital splits between HPC and IPC under each scenario.

The final long panel is sorted by `company` and `month`.

4. Interpretation of the Haifa K series

4.1 Units and levels

All K series in `20_K_cluster_Haifa_HPC_IPC_monthly.tsv` are:

- In **thousands of NIS**.
- **Deflated to 2019 prices**, using an OECD capital deflator.
- At **monthly frequency**, from December 2018 through December 2024.

So, for example, a value of `K_central = 12_000` for the cluster means:

The Haifa cluster (HPC + IPC) has a real capital stock equivalent to 12 billion NIS in 2019 prices.

4.2 Scenario meaning: low / central / high

The three K columns `K_low`, `K_central`, and `K_high` represent different assumptions about the **economic depreciation rate** δ :

- **Low δ scenario** (`K_low`):
 - Assets are assumed to depreciate **slowly**.
 - Capital stocks are therefore **higher** and more persistent.
- **Central δ scenario** (`K_central`):
 - This is the **benchmark** or “best guess” about depreciation.
 - It is the main series used in baseline regressions.
- **High δ scenario** (`K_high`):
 - Assets are assumed to depreciate **quickly**.
 - Capital stocks are **lower** and decline more rapidly over time.

Empirically, you observe the expected ordering:

$$K_{\text{low}}(t) > K_{\text{central}}(t) > K_{\text{high}}(t)$$

for both HPC and IPC, and thus for the cluster as well.

This structure allows you to do **robustness checks** in the econometric work by re-estimating key models under alternative depreciation rates.

4.3 Operator vs landlord vs cluster K

The panel allows several distinct perspectives:

1. **Operator K (HPC)** – `company == "Haifa Port Company (legacy)"`
2. Captures the capital stock directly controlled by the legacy operator.
3. Includes equipment and structures that remain on HPC's balance sheet, constructed via the lumpy Path B process.
4. **Landlord K (IPC)** – `company == "Israel Ports Company (Haifa cluster)"`
5. Captures the capital stock owned by the national port authority (breakwaters, berths, major marine infrastructure, some yards, etc.) and allocated to Haifa.

6. Built via smooth PIM-lin from IPC's Haifa-specific PPE tables.

7. **Cluster K (HPC + IPC)** – `company == "Haifa cluster (HPC + IPC)"`

8. The sum of operator and landlord capital.

9. This is a key object in throughput and capacity models (e.g. $\ln(\text{TEU})$ on $\ln(K)$), since it more closely approximates the **total physical capital** deployed in the Haifa container cluster.

By comparing `K_HPC_*` and `K_IPC_*` over time, you can analyze how the **ownership composition** of port capital evolves (e.g. shifts from operator to landlord) and how this interacts with the reforms.

5. How this K panel enters the econometric models

The Haifa K panel is designed to plug directly into the thesis' econometric models.

1. Model 1B – Capital deepening and labor productivity

2. You can construct variables like $\ln(K/L)$ or $\ln(K)$ at the **operator level**:

- Use `company == "Haifa Port Company (legacy)"` and the `K_central` series, combined with a labor-input series (e.g. hours).

3. You can also use cluster-level K to control for the total capital environment at Haifa.

4. Model 2 – Throughput / capacity models

5. Use **cluster K** (`company == "Haifa cluster (HPC + IPC)"`) as a measure of total physical capacity in $\ln(\text{TEU})$ regressions.

6. The three scenarios `K_low`, `K_central`, `K_high` can be used for robustness on capacity measurement.

7. Dynamic and distributional analysis

8. The panel structure allows you to analyze how K evolves around key events (e.g. Bayport opening, HPC privatization) and how ownership shifts between HPC and IPC.

9. Because the series are monthly, you can align capital changes with monthly throughput and (eventually) labor hours.

6. Summary

To summarize:

- The Haifa K series are built from **company-level financial statements** (HPC and IPC), deflated to 2019-NIS, and transformed into **annual PIM capital stocks** under multiple depreciation scenarios.

- For IPC, you apply a **smooth monthly PIM-lin** procedure to obtain monthly capital under each scenario.
- For HPC, you construct a richer **Path B** series by combining a lumpy big-project module with a smooth background capital component that matches the annual PIM stocks for each scenario.
- You then merge the operator and landlord series into a **Haifa cluster K panel** with three company values (HPC, IPC, cluster) and three scenarios (low, central, high), ensuring that for every month and scenario:

$$K_{\text{cluster},s}(t) = K_{\text{HPC},s}(t) + K_{\text{IPC},s}(t).$$

This panel is the backbone for all Haifa-related capital measures in the thesis, and it is constructed to be transparent, consistent with the accounting data, and flexible enough to support multiple depreciation assumptions and econometric specifications.